Remove all duplicates and clean my code

1 message

Pranav Madhu <pranavmadhu8055@gmail.com> Sat, 24 Oct 2026 at 12:16 pm
Draft

```
// Automatic FlutterFlow imports
import '/backend/schema/structs/index.dart';
import '/flutter_flow/ff_builtin_enums.dart';
import '/actions/actions.dart' as action_blocks;
import "package:custom_code_widgets_jf41ty/backend/schema/structs/index.dart"
    as custom_code_widgets_jf41ty_data_schema;
import "package:n_e_w_a_u_d_i_t_library_4x39ut/backend/schema/structs/index.dart"
    as n_e_w_a_u_d_i_t_library_4x39ut_data_schema;
import "package:iggl_irregularityupdation_jo6j0q/backend/schema/structs/index.dart"
    as iggl_irregularityupdation_jo6j0q_data_schema;
import "package:others_4a3ytl/backend/schema/structs/index.dart"
    as others_4a3ytl_data_schema;
import "package:custom_code_widgets_jf41ty/backend/schema/enums/enums.dart"
    as custom_code_widgets_jf41ty_enums;
import 'package:ff_theme/flutter_flow/flutter_flow_theme.dart';
import '/flutter_flow/flutter_flow_util.dart';
import 'index.dart'; // Imports other custom widgets
import '/custom_code/actions/index.dart'; // Imports custom actions
import '/flutter_flow/custom_functions.dart'; // Imports custom functions
import 'package:flutter/material.dart';
// Begin custom widget code
// DO NOT REMOVE OR MODIFY THE CODE ABOVE!

import 'package:flutter/services.dart';
import 'package:custom_code_library_rtycn8/app_state.dart'
    as custom_code_library_rtycn8_app_state;

import 'dart:math';
import 'package:http/http.dart' as http;
import 'dart:convert';
import 'package:intl/intl.dart';

class AlcForm extends StatefulWidget {
  final AlcInput input;
  final ApiService apiService;
  final Map<int, String> itemTypes;

  const AlcForm({
    super.key,
    required this.input,
    required this.apiService,
    required this.itemTypes,
  });

  @override
  State<AlcForm> createState() => _AlcFormState();
}

class _AlcFormState extends State<AlcForm> {
  late List<AlcRow> _rows;
  double _totGw = 0.0;
  double _totStwt = 0.0;
  double _totNetwt = 0.0;
  bool _isLoading = false;
  bool _showAutoCalc = false;
  bool _showAddRow = false;
  bool _hideItemCount = false;

  @override
  void initState() {
    super.initState();
    _rows = [];
    _setupVisibility();
    _initFirstRow();
  }

  void _setupVisibility() {
    final s = widget.input.statusMain;
    final s1 = widget.input.statusSub;
```

```
  if ([2, 5, 6, 12, 13, 27, 28].contains(s)) {
    if (![23, 9, 10].contains(s1)) {
      _showAutoCalc = true;
    }
  }
  if ([29, 30, 31].contains(s) || [29, 30, 31, 38].contains(s1)) {
    _showAddRow = true;
    _showAutoCalc = true;
    _hideItemCount = true;
  }
}

Future<void> _initFirstRow() async {
  setState(() => _isLoading = true);
  try {
    final options = await widget.apiService.fetchPurityParams(
        widget.input.pledgeNo,
        widget.input.brid.toString(),
        widget.input.typeId,
        widget.input.scheme,
        widget.input.pledgeDate);
    _rows.add(AlcRow(
      itemCount: _hideItemCount ? 0 : widget.input.dbItemCount,
      gw: 0.0,
      stwt: 0.0,
      purityText: '',
      estimate: 0.0,
      netwt: 0.0,
      autoLoss: 0.0,
      typeId: widget.input.typeId,
      purityOptions: options,
      cntwt: widget.input.cntwt,
    ));
  } finally {
    setState(() => _isLoading = false);
  }
}

Future<void> _addRow() async {
  setState(() => _isLoading = true);
  try {
    final options = await widget.apiService.fetchPurityParams(
        widget.input.pledgeNo,
        widget.input.brid.toString(),
        widget.input.typeId,
        widget.input.scheme,
        widget.input.pledgeDate);
    setState(() {
      _rows.add(AlcRow(
        itemCount: 0,
        gw: 0.0,
        stwt: 0.0,
        purityText: '',
        estimate: 0.0,
        netwt: 0.0,
        autoLoss: 0.0,
        typeId: widget.input.typeId,
        purityOptions: options,
        cntwt: 0.0,
      ));
    });
  } finally {
    setState(() => _isLoading = false);
  }
}

double remain_netwt(double rgwt, double rstwt) {
  double nwt = ((rgwt - rstwt) * widget.input.deduRate / 100);
  return double.parse(((rgwt - rstwt) - nwt).toStringAsFixed(2));
}

Future<void> _showAlert(String message) async {
  await showDialog(
    context: context,
    builder: (ctx) => AlertDialog(
      title: const Text('Alert'),
      content: Text(message),
      actions: [
        TextButton(
            onPressed: () => Navigator.pop(ctx), child: const Text('OK')),
      ],
    ),
  );
}
```

```
double _getNumericPurityFromHeaderText(String purityText) {
  if (purityText.isEmpty) return 0.0;
  List<String> purity = purityText.split('~');
  if (purity.isEmpty) return 0.0;

  final header = purity[0];
  List<String> spurity = [];

  if (header == "Purity Above 85(Above 20.5-ct)") {
    spurity = header.split('(');
    if (header.length >= 15) {
      spurity.add(header.substring(13, 15));
    } else {
      spurity.add("85");
    }
  } else if ([
    "Purity-70(18-ct)",
    "Purity-75(19-ct)",
    "Purity-80(20-ct)",
    "Purity-85(20.5-ct)"
  ].contains(header)) {
    spurity = header.split('(')[0].split('-');
  } else if (header.startsWith("Purity>=")) {
    spurity = header.split('=');
  } else {
    spurity = header.split('-');
    if (spurity.length == 1) {
      spurity = header.split('=');
    }
  }

  if (spurity.length > 1) {
    return double.tryParse(spurity[1].replaceAll(')', '').trim()) ?? 0.0;
  }
  return 0.0;
}

Future<void> _autoCalculate() async {
  await _recalculate();
  double currentGwSum = 0.0;
  double currentStSum = 0.0;
  double currentNetSum = 0.0;

  for (var row in _rows) {
    if (row.purityText.isNotEmpty && row.purityText.contains('~')) {
      currentGwSum += row.gw;
      currentStSum += row.stwt;
      currentNetSum += row.netwt;
    }
  }

  double remGw = widget.input.cgwt - currentGwSum;
  double remSt = widget.input.cstwt - currentStSum;
  double extraNet = remain_netwt(remGw, remSt);

  setState(() {
    _totGw = widget.input.cgwt;
    _totStwt = widget.input.cstwt;
    _totNetwt = currentNetSum + extraNet;
  });

  _showAlert("Auto Calculated. Remaining Net Added: $extraNet");
}

Future<void> _recalculate() async {
  double gwSum = 0.0;
  double stSum = 0.0;
  double netSum = 0.0;

  final rowsCopy = List<AlcRow>.from(_rows);
  for (var row in rowsCopy) {
    if (row.purityText.isEmpty || !row.purityText.contains('~')) {
      row.netwt = 0.0;
      row.estimate = 0.0;
      row.autoLoss = 0.0;
      continue;
    }

    List<String> purity = row.purityText.split('~');
    double purityVal =
        (purity.length > 1) ? (double.tryParse(purity[1]) ?? 0.0) : 0.0;

    if (purityVal >= 20 && purityVal <= 65) {
```

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      row.estimate = (100 - ((purityVal / 87) * 100)).roundToDouble();
    } else {
      row.estimate = purityVal;
    }

    double factor = (row.estimate / 100 == 0) ? 1.0 : (row.estimate / 100);
    double clnt = ((row.gw - row.stwt) * factor);

    if (row.estimate == 0) {
      row.netwt = double.parse(clnt.toStringAsFixed(2));
    } else {
      row.netwt =
          double.parse(((row.gw - row.stwt) - clnt).toStringAsFixed(2));
    }

    if (row.netwt < 0) {
      row.netwt = 0;
      await _showAlert("Net weight cannot be negative");
    }

    double ilv = (row.cntwt * widget.input.lendRate).roundToDouble();
    double ialv = (row.netwt * widget.input.lendRate).roundToDouble();
    row.autoLoss = ilv - ialv;
  }

  for (var row in _rows) {
    gwSum += row.gw;
    stSum += row.stwt;
    netSum += row.netwt;
  }

  setState(() {
    _totGw = gwSum;
    _totStwt = stSum;
    _totNetwt = netSum;
  });
}

Future<void> _onItemCountChanged(AlcRow row, String value) async {
  int val = int.tryParse(value) ?? 0;
  row.itemCount = val;

  int totalCount = _rows.fold(0, (sum, r) => sum + r.itemCount);
  int dbCount = widget.input.dbItemCount;

  if (dbCount - totalCount >= 1) {
    await _addRow();
  } else if (dbCount - totalCount < 0) {
    await _showAlert("Item count exceeds actual Items");
    int otherRowsCount = totalCount - val;
    row.itemCount = dbCount - otherRowsCount;
    if (row.itemCount < 0) row.itemCount = 0;
  }
  setState(() {});
}

void _onStoneChanged(AlcRow row, String value) {
  double val = double.tryParse(value) ?? 0.0;
  row.stwt = val;

  if (row.stwt > row.gw) {
    row.stwt = 0;
    _showAlert("Stone Weight exceeds Gross weight");
  } else {
    double currentStSum = _rows.fold(0.0, (sum, r) => sum + r.stwt);
    int irreg = widget.input.statusMain;
    if (![9, 10, 23].contains(irreg) && widget.input.cstwt < currentStSum) {
      row.stwt = 0;
      _showAlert("Stone Weight exceeds pledge Stone weight");
    }
  }
  _recalculate();
}

Future<void> _onTypeChanged(AlcRow row, int? newTypeId) async {
  if (newTypeId == null) return;
  setState(() => _isLoading = true);
  try {
    row.typeId = newTypeId;
    row.purityText = '';
    row.purityOptions = await widget.apiService.fetchPurityParams(
        widget.input.pledgeNo,
        widget.input.brid.toString(),
        newTypeId,
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          widget.input.scheme,
          widget.input.pledgeDate);
      _recalculate();
    } finally {
      setState(() => _isLoading = false);
    }
  }

  void _handleSave() {
    bool allFilled = _rows.every((row) => row.gw > 0 && row.purityText != '');

    if (!allFilled) {
      _showAlert("Please Enter Values or Delete Rows to Close Form");
      return;
    }

    String subrecord = _rows.map((row) {
      double pVal = _getNumericPurityFromHeaderText(row.purityText);
      return "${widget.input.itemId}*${row.itemCount}*${row.gw}*${row.stwt}*${row.netwt}*${pVal.
toString()}*${row.autoLoss}";
    }).join('!');

    Navigator.pop(
        context,
        AlcResult(
          tGrwt: _totGw,
          tStwt: _totStwt,
          tNtwt: _totNetwt,
          subrecord: subrecord,
        ));
  }

  @override
  Widget build(BuildContext context) {
    final size = MediaQuery.of(context).size;
    return AlertDialog(
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
      backgroundColor: Colors.white,
      title: Row(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children: [
          Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              const Text('Item Details',
                  style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold)),
              Text(widget.input.itemName,
                  style: const TextStyle(
                      fontSize: 13,
                      fontWeight: FontWeight.w500,
                      color: Colors.grey)),
            ],
          ),
          IconButton(
            icon: const Icon(Icons.close),
            onPressed: () => Navigator.pop(context),
          ),
        ],
      ),
      content: SizedBox(
        width: size.width * 0.9,
        height: size.height * 0.8,
        child: Column(
          children: [
            if (_isLoading) const LinearProgressIndicator(),
            Container(
              decoration: BoxDecoration(
                color: Colors.white,
                borderRadius: BorderRadius.circular(12),
                boxShadow: [
                  BoxShadow(
                    color: Colors.black.withOpacity(0.1),
                    blurRadius: 8,
                    offset: const Offset(0, 4),
                  ),
                ],
              ),
              padding: const EdgeInsets.all(20.0),
              child: Row(
                mainAxisAlignment: MainAxisAlignment.spaceAround,
                children: [
                  _HeaderStat('Total GW', _totGw.toStringAsFixed(3)),
                  _HeaderStat('Total Stone', _totStwt.toStringAsFixed(3)),
                  _HeaderStat('Total Net', _totNetwt.toStringAsFixed(3),
```

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            isPrimary: true),
      ],
    ),
  ),
  Expanded(
    child: Container(
      margin: const EdgeInsets.only(top: 20),
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(12),
        boxShadow: [
          BoxShadow(
            color: Colors.black.withOpacity(0.1),
            blurRadius: 10,
            offset: const Offset(0, 5),
          ),
        ],
      ),
      clipBehavior: Clip.antiAlias,
      child: SingleChildScrollView(
        scrollDirection: Axis.vertical,
        child: SingleChildScrollView(
          scrollDirection: Axis.horizontal,
          child: DataTable(
            columnSpacing: 20,
            headingRowColor:
                MaterialStateProperty.all(const Color(0xFF667eea)),
            headingTextStyle: const TextStyle(
                fontWeight: FontWeight.bold, color: Colors.white),
            columns: const [
              DataColumn(
                  label: Text('Item',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('No of pieces',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Gross Wt',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Net wt',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Stone Wt',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Cur.Purity',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Item Type',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Purity',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Purity Deduction',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
              DataColumn(
                  label: Text('Delete',
                      style: TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 12))),
            ],
            rows: List.generate(_rows.length, (index) {
              final row = _rows[index];
              return DataRow(
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key: ObjectKey(row),
cells: [
  DataCell(Text(widget.input.itemName,
      style: const TextStyle(fontSize: 12))),
  DataCell(SizedBox(
    width: 60,
    child: TextFormField(
      initialValue: row.itemCount.toString(),
      keyboardType: TextInputType.number,
      enabled: !_hideItemCount,
      onChanged: (v) => _onItemCountChanged(row, v),
      inputFormatters: [
        FilteringTextInputFormatter.allow(
            RegExp(r'^\d*\.?\d*'))
      ],
      decoration: const InputDecoration(
          isDense: true,
          border: OutlineInputBorder()),
    ),
  )),
  DataCell(SizedBox(
    width: 80,
    child: TextFormField(
      decoration: const InputDecoration(
          isDense: true,
          border: OutlineInputBorder()),
      initialValue: row.gw.toString(),
      keyboardType:
          const TextInputType.numberWithOptions(
              decimal: true),
      inputFormatters: [
        FilteringTextInputFormatter.allow(
            RegExp(r'^\d*\.?\d*'))
      ],
      onChanged: (v) {
        row.gw = double.tryParse(v) ?? 0.0;
        _recalculate();
      },
    ),
  )),
  DataCell(Text(row.netwt.toStringAsFixed(3))),
  DataCell(SizedBox(
    width: 80,
    child: TextFormField(
      decoration: const InputDecoration(
          isDense: true,
          border: OutlineInputBorder()),
      initialValue: row.stwt.toString(),
      inputFormatters: [
        FilteringTextInputFormatter.allow(
            RegExp(r'^\d*\.?\d*'))
      ],
      keyboardType:
          const TextInputType.numberWithOptions(
              decimal: true),
      onChanged: (v) => _onStoneChanged(row, v),
    ),
  )),
  DataCell(Text(widget.input.curPurity,
      style: const TextStyle(fontSize: 12))),
  DataCell(SizedBox(
    width: 120,
    child: DropdownButtonFormField<int>(
      decoration: const InputDecoration(
          isDense: true,
          border: OutlineInputBorder()),
      hint: const Text('Select Type',
          style: TextStyle(fontSize: 12)),
      value: widget.itemTypes.containsKey(row.typeId)
          ? row.typeId
          : null,
      isExpanded: true,
      items: widget.itemTypes.entries
          .map((e) => DropdownMenuItem(
              value: e.key,
              child: Text(e.value,
                  overflow: TextOverflow.ellipsis,
                  style:
                      const TextStyle(fontSize: 12))))
          .toList(),
      onChanged: (v) => _onTypeChanged(row, v),
    ),
  )),
  DataCell(SizedBox(
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                                width: 150,
                                child: DropdownButtonFormField<String>(
                                  decoration: const InputDecoration(
                                      isDense: true,
                                      border: OutlineInputBorder()),
                                  hint: const Text('Select Purity',
                                      style: TextStyle(fontSize: 12)),
                                  value:
                                      row.purityOptions.contains(row.purityText)
                                          ? row.purityText
                                          : null,
                                  isExpanded: true,
                                  items: row.purityOptions
                                      .map((e) => DropdownMenuItem(
                                            value: e,
                                            child: Text(e.split('~').first,
                                                overflow: TextOverflow.ellipsis,
                                                style: const TextStyle(
                                                    fontSize: 13)),
                                          ))
                                      .toList(),
                                  onChanged: (v) {
                                    if (v != null) {
                                      setState(() {
                                        row.purityText = v;
                                        _recalculate();
                                      });
                                    }
                                  },
                                ),
                              )),
                              DataCell(Text(row.estimate.toStringAsFixed(2))),
                              DataCell(IconButton(
                                icon: const Icon(Icons.delete, color: Colors.red),
                                onPressed: () {
                                  setState(() {
                                    _rows.removeAt(index);
                                    _recalculate();
                                  });
                                },
                              )),
                            ],
                          );
                        }),
                      ),
                    ),
                  ),
                ),
              )
            ],
          ),
        ),
        actions: [
          if (_showAutoCalc)
            TextButton.icon(
              onPressed: _autoCalculate,
              label: const Text('Auto Calc'),
              icon: const Icon(Icons.calculate),
            ),
          if (_showAddRow)
            TextButton.icon(
              onPressed: _addRow,
              label: const Text('Add Row'),
              icon: const Icon(Icons.add),
            ),
          ElevatedButton.icon(
            icon: const Icon(Icons.check),
            label: const Text('Save'),
            onPressed: _handleSave,
          ),
        ],
      );
    }
  }

  class AlcRow {
    int itemCount;
    double gw;
    double stwt;
    String purityText;
    double estimate;
    double netwt;
    double autoLoss;
    int typeId;
```

```
  List<String> purityOptions;
  double cntwt;

  AlcRow({
    required this.itemCount,
    required this.gw,
    required this.stwt,
    required this.purityText,
    required this.estimate,
    required this.netwt,
    required this.autoLoss,
    required this.typeId,
    this.purityOptions = const [],
    required this.cntwt,
  });
}

class _HeaderStat extends StatelessWidget {
  final String label;
  final String value;
  final bool isPrimary;
  final Color? textColor;
  const _HeaderStat(this.label, this.value,
      {this.isPrimary = false, this.textColor});

  @override
  Widget build(BuildContext context) {
    return Column(
      children: [
        Text(label,
            style: Theme.of(context)
                .textTheme
                .bodySmall
                ?.copyWith(color: textColor?.withOpacity(0.9))),
        Text(value,
            style: Theme.of(context).textTheme.titleMedium?.copyWith(
                  fontWeight: FontWeight.bold,
                  color: textColor ??
                      (isPrimary
                          ? Theme.of(context).colorScheme.primary
                          : null),
                )),
      ],
    );
  }
}

class PacketDemoPage extends StatefulWidget {
  final double? width;
  final double? height;

  const PacketDemoPage({
    super.key,
    this.width,
    this.height,
  });

  @override
  State<PacketDemoPage> createState() => _PacketDemoPageState();
}

class _PacketDemoPageState extends State<PacketDemoPage> {
  late AuditState audit;
  bool _isLoading = false;
  Map<int, String> _statusOptions = {};
  Map<int, String> _itemTypes = {};
  List<List<String>> _purityOptionsList = [];
  bool isEnabled = true;

  @override
  void initState() {
    super.initState();
    FFAppState().selectbrid =
        custom_code_library_rtycn8_app_state.FFAppState().SelectedBranchID;
    FFAppState().cmbStatus = 0;
    FFAppState().cmbStatus2 = 0;
    FFAppState().loss = '';

    audit = AuditState(
      lendRate: 0,
      pledgeBalance: 0,
      totalNetWtDb: 0.0,
      scheme: '',
      actweight: 0.0,
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    netweight: 0.0,
    pledgeDate: DateTime(2014, 1, 1),
    marketRate: 3100,
  );

  _loadIrr();
}

Future<void> _loadIrr() async {
  setState(() => _isLoading = true);
  try {
    final api = ApiService();
    final statuses = await api.fetchStatusOptions(FFAppState().selectbrid, 90);
    setState(() {
      _statusOptions = statuses;
      if (!_statusOptions.containsKey(FFAppState().cmbStatus)) {
        if (_statusOptions.containsKey(0)) {
          FFAppState().cmbStatus = 0;
        } else if (_statusOptions.isNotEmpty) {
          FFAppState().cmbStatus = _statusOptions.keys.first;
        }
      }
    });
  } finally {
    setState(() => _isLoading = false);
  }
}

DateTime? _parsePledgeDate(String raw) {
  if (raw.isEmpty) return null;
  try {
    final dateStr = raw.trim().replaceAll(RegExp(r'[./]'), '-');
    final parts = dateStr.split('-');
    if (parts.length != 3) return null;

    final day = parts[0].padLeft(2, '0');
    final monRaw = parts[1];
    final year = parts[2];
    final month = (monRaw.isEmpty)
        ? monRaw
        : monRaw[0].toUpperCase() + monRaw.substring(1).toLowerCase();

    final normalized = '$day-$month-$year';
    final df = DateFormat('dd-MMM-yyyy', 'en_US');
    return df.parseStrict(normalized);
  } catch (e) {
    debugPrint('Date parse error: $e');
    return null;
  }
}

Future<void> _populatePurityOptions(List<Result1Struct> items, ApiService api) async {
  _purityOptionsList.clear();
  final typeCache = <int, List<String>>{};

  for (final item in items) {
    final typeId = item.colTypid ?? 0;
    if (!typeCache.containsKey(typeId)) {
      typeCache[typeId] = await api.fetchPurityParams(
          FFAppState().pledgeNumIrr,
          FFAppState().selectbrid,
          typeId,
          audit.scheme,
          audit.pledgeDate);
    }
    final options = typeCache[typeId] ?? [];
    _purityOptionsList.add(options);

    String purityText = item.colPurity ?? '';
    if (purityText.isNotEmpty && options.isNotEmpty) {
      if (!options.contains(purityText)) {
        final match = options.firstWhere(
          (opt) => opt.startsWith(purityText),
          orElse: () => '',
        );
        if (match.isNotEmpty) item.colPurity = match;
      }
    }
  }
}

Future<void> _loadData() async {
  setState(() => _isLoading = true);
```

```
  try {
    final api = ApiService();
    final items = await api.fetchPledgeItems(
        FFAppState().pledgeNumIrr, FFAppState().selectbrid);

    _itemTypes = await api.fetchItemTypes();
    final mRate = await api.fetchMarketRate();
    final pBal = await api.fetchPledgeBalance(FFAppState().pledgeNumIrr);
    final invDetails = await api.fetchInventoryDetails(
        FFAppState().selectbrid, FFAppState().pledgeNumIrr);

    double lRate = 0.0;
    if (invDetails != null) {
      lRate = double.tryParse(invDetails['LND_RATE']?.toString() ?? '0') ?? 0.0;
      audit.scheme = invDetails['SCHEME_NM']?.toString() ?? '';
      audit.actweight = double.tryParse(invDetails['ACTUAL_GROSS_WT']?.toString() ?? '0') ?? 0.0;
      audit.netweight = double.tryParse(invDetails['NET_WEIGHT']?.toString() ?? '0') ?? 0.0;

      final parsedDate = _parsePledgeDate(invDetails['PLEDGEDT']?.toString() ?? '');
      if (parsedDate != null) {
        audit.pledgeDate = parsedDate;
      }
    } else {
      lRate = await api.fetchLendRate(FFAppState().pledgeNumIrr);
    }

    await _populatePurityOptions(items, api);

    setState(() {
      FFAppState().Dtview = items;
      audit.marketRate = mRate;
      audit.lendRate = lRate;
      audit.pledgeBalance = pBal;
    });
    _recalculateTotals();
  } finally {
    setState(() => _isLoading = false);
  }
}

void _showMessage(String msg) {
  showDialog(
    context: context,
    builder: (ctx) => AlertDialog(
      title: const Text('Alert'),
      content: Text(msg),
      actions: [
        TextButton(
            onPressed: () => Navigator.pop(ctx), child: const Text('OK'))
      ],
    ),
  );
}

Future<AlcResult?> _openAlcForm(AlcInput input) async {
  return showDialog<AlcResult>(
    context: context,
    barrierDismissible: false,
    builder: (ctx) => AlcForm(
      input: input,
      apiService: ApiService(),
      itemTypes: _itemTypes,
    ),
  );
}

Future<void> _dtviewCellEndEdit(int rowIndex, int columnIndex) async {
  if (FFAppState().Dtview.isNotEmpty) {
    isEnabled = false;

    if (columnIndex > 2) {
      double loss = 0;
      double stone = 0;
      double nt = 0;
      double clnt = 0;
      double gw = 0;
      int rowCount = FFAppState().Dtview.length;

      List<int> iloanvalue = List.filled(rowCount, 0);
      List<int> iauloanvalue = List.filled(rowCount, 0);
      int totloanvalue = 0;
      int aalv = 0;
      int totaalv = 0;
```

```
for (int i = 0; i < rowCount; i++) {
  if (columnIndex == 8 &&
      (FFAppState().Dtview[i].colPurity ?? '').isNotEmpty) {
    var row = FFAppState().Dtview[i];

    if ((row.colGw ?? '').isNotEmpty &&
        (row.colPurity ?? '').isNotEmpty) {
      List<String> purity = row.colPurity!.split('~');
      double pVal = purity.length > 1 ? double.tryParse(purity[1]) ?? 0 : 0;

      if (pVal >= 20 && pVal <= 65) {
        row.colEstimat = (100 - ((pVal / 87) * 100)).roundToDouble().toString();
      } else if (pVal == 100) {
        row.colEstimat = pVal.toString();
      } else {
        row.colEstimat = pVal.toString();
      }

      double gwVal = double.tryParse(row.colGw ?? '0') ?? 0;
      double stVal = double.tryParse(row.colStwt ?? '0') ?? 0;
      double estVal = double.tryParse(row.colEstimat ?? '0') ?? 0;

      clnt = ((gwVal - stVal) * ((estVal / 100 == 0) ? 1 : estVal / 100));

      if (estVal == 0) {
        row.colNetwt = clnt.toString();
      } else {
        if (_isStonePurity_Deductible() &&
            stVal > 0 &&
            estVal != 100.0) {
          List<String> ppurity = row.colPurity!.split('~');
          List<String> spurity = ppurity[0].split('-');

          if (ppurity[0] == "Purity Above 85(Above 20.5-ct)") {
            spurity = ppurity[0].split('(');
            if (spurity[0].length >= 15) {
              spurity.add(spurity[0].substring(13, 15));
            }
          } else if ([
            "Purity-70(18-ct)",
            "Purity-75-(19-ct)",
            "Purity-70-(17-ct)",
            "Purity-75-(18-ct)",
            "Purity-79-(19-ct)",
            "Purity-83-(20-ct)",
            "Purity-87-(21-ct)",
            "Purity-91.6-Above(22-ct)",
            "Purity-80(20-ct)",
            "Purity-85(20.5-ct)"
          ].contains(ppurity[0])) {
            spurity = ppurity[0].split('(')[0].split('-');
          } else if (spurity.length != 2) {
            spurity = spurity[0].split('-');
          }
          if (spurity.length == 1) {
            spurity = ppurity[0].split('=');
          }

          double spurityVal = spurity.length > 1 ? double.tryParse(spurity[1]) ?? 0 : 0;
          String itemName = FFAppState().Dtview[rowIndex].colItem ?? '';

          if (spurityVal > 70) {
            row.colNetwt = await _stonePurityDeduction(
                ppurity.length > 1 ? ppurity[1] : '0',
                row.colGw ?? '0',
                row.colStwt ?? '0',
                itemName);
          } else {
            row.colNetwt = await _stonePurityDeduction(
                row.colEstimat ?? '0',
                row.colGw ?? '0',
                row.colStwt ?? '0',
                itemName);
          }
          _showMessage("Stone wise Purity Deduction calculated");
        } else {
          row.colNetwt = double.parse(((gwVal - stVal) - clnt).toStringAsFixed(2)).toString();
        }
      }

      nt = nt + (double.tryParse(row.colNetwt ?? '0') ?? 0);
      FFAppState().totalNetWtVar = double.parse(nt.toStringAsFixed(2));

      if ([36, 37, 38].contains(FFAppState().cmbStatus)) {
```

```
          double cntWt = row.colCntwt ?? 0;
          double lndRate = audit.lendRate;
          iloanvalue[i] = double.parse((cntWt * lndRate).toStringAsFixed(2)).round();
          iauloanvalue[i] = (double.tryParse(row.colNetwt ?? '0')! * lndRate).round();
          totloanvalue = (audit.netweight * lndRate).toInt();
          row.colAutoloss = (iloanvalue[i] - iauloanvalue[i]).toString();

          int totilv = 0;
          int totialv = 0;
          for (int index = 0; index < rowCount; index++) {
            if (iloanvalue[index] != 0) {
              totilv += iloanvalue[index];
              totialv += iauloanvalue[index];
            }
          }
          aalv = totloanvalue - totilv;
          totaalv = (aalv == -1 ? 0 : aalv) + totialv;
          double pBal = audit.pledgeBalance.toDouble();
          double val = pBal - totaalv <= -1 ? 0 : pBal - totaalv;

          FFAppState().update(() {
            FFAppState().txtLoss = val;
          });
        }

        if ([5, 20, 6, 29, 30, 12, 13, 27, 28, 2, 1]
                .contains(FFAppState().cmbStatus) &&
            FFAppState().cmbStatus2 == 0) {
          double cntWt = row.colCntwt ?? 0;
          double lndRate = audit.lendRate;
          iloanvalue[i] = (cntWt * lndRate).round();
          iauloanvalue[i] = (double.tryParse(row.colNetwt ?? '0.0')! * lndRate).round();
          totloanvalue = (audit.netweight * lndRate).toInt();
          row.colAutoloss = (iloanvalue[i] - iauloanvalue[i]).toString();

          int totilv = 0;
          int totialv = 0;
          for (int index = 0; index < rowCount; index++) {
            if (iloanvalue[index] != 0) {
              totilv += iloanvalue[index];
              totialv += iauloanvalue[index];
            }
          }
          aalv = totloanvalue - totilv;
          totaalv = aalv + totialv;

          double pBal = audit.pledgeBalance.toDouble();
          if (pBal > totaalv) {
            double val = pBal - totaalv;
            FFAppState().update(() {
              FFAppState().txtLoss = val;
            });
          } else {
            FFAppState().update(() {
              FFAppState().loss = "0";
              FFAppState().txtLoss = 0.0;
            });
          }
        }
      }
    } else if (columnIndex == 5) {
      stone = stone + (double.tryParse(FFAppState().Dtview[i].colStwt ?? '0') ?? 0);
      FFAppState().totalStoneVar = stone;
    } else if (columnIndex == 3) {
      gw = gw + (double.tryParse(FFAppState().Dtview[i].colGw ?? '0') ?? 0);
      FFAppState().totalGwVar = gw;
    }

    if (FFAppState().cmbStatus == 4 && columnIndex == 5) {
      double rt = audit.marketRate;
      FFAppState().Dtview[rowIndex].colNetwt = "0.0";
      double totGw = FFAppState().totalGwVar;
      double totSt = FFAppState().totalStoneVar;
      double val = ((totGw - totSt) * rt);
      FFAppState().update(() {
        FFAppState().loss = val.toString();
        FFAppState().txtLoss = val;
      });
    }
  }
} else {
  var row = FFAppState().Dtview[rowIndex];

  bool isSpuriousOrLowQuality = [5, 6, 29, 30, 31, 12, 13, 27, 28, 2, 38]
```

```
            .contains(FFAppState().cmbStatus);
      bool isSpecialStatus = [23, 9, 10].contains(FFAppState().cmbStatus);

      if ((row.colStatus ?? false) == true &&
          ((FFAppState().cmbStatus2 != 1 && FFAppState().cmbStatus2 > 0 && isSpuriousOrLowQuality) ||
           isSpecialStatus)) {

        var res = await _openAlcForm(AlcInput(
            itemName: row.colItem ?? '',
            deduRate: row.colDeducrate.toDouble(),
            dbItemCount: row.colDbitemcount ?? 0,
            itemId: row.colItemid ?? 0,
            brid: int.tryParse(FFAppState().selectbrid) ?? 0,
            pledgeNo: FFAppState().pledgeNumIrr,
            cgwt: row.colCgwt ?? 0.0,
            cstwt: row.colCstwt.toDouble(),
            statusMain: FFAppState().cmbStatus,
            cntwt: row.colCntwt ?? 0.0,
            scheme: audit.scheme,
            pledgeDate: audit.pledgeDate,
            statusSub: isSpecialStatus ? 0 : FFAppState().cmbStatus2,
            typeId: row.colTypid ?? 0,
            curPurity: row.colCurpurity ?? '',
            itemType: '',
            lendRate: audit.lendRate));

        if (res != null) {
          if (!isSpecialStatus && res.tGrwt > (row.colCgwt ?? 0.0)) {
            _showMessage("Calculated Item Weight Exceeds Actual Gross Weight. Please Calculate Again");
            row.colStatus = false;
            row.colItemcount = "";
            row.colGw = "";
            row.colStwt = "";
            row.colNetwt = "";
            row.colSubrecord = "";
            return;
          }
          row.colItemcount = (row.colDbitemcount ?? 0).toString();
          row.colGw = res.tGrwt.toString();
          row.colStwt = res.tStwt.toString();
          row.colNetwt = res.tNtwt.toString();
          row.colSubrecord = res.subrecord;

          if (_isStonePurity_Deductible() &&
              (double.tryParse(row.colStwt ?? '0') ?? 0) > 0) {
            row.colNetwt = await _alcStonePurityDeduction(
                row.colNetwt!, row.colGw!, row.colStwt!, row.colItem ?? '');
          }
          _recalculate();
        }
      }
    }
  }
  setState(() {});
}

Future<void> _handleStatusChange(int? status) async {
  if (status == null) return;

  setState(() {
    FFAppState().cmbStatus = status;
    FFAppState().selectbrid = custom_code_library_rtycn8_app_state.FFAppState().SelectedBranchID;
    FFAppState().loss = '0.0';
    FFAppState().txtLoss = 0.0;
  });

  await _loadData();

  setState(() {
    FFAppState().totalGwVar = 0.0;
    FFAppState().totalNetWtVar = 0.0;
    FFAppState().totalStoneVar = 0.0;

    final autoLossStatuses = [3, 7, 15, 16, 17, 18, 19, 21, 24, 25, 26, 32];

    if (autoLossStatuses.contains(FFAppState().cmbStatus)) {
      if (FFAppState().cmbStatus == 7) {
        FFAppState().loss = audit.pledgeBalance.toString();
      } else if (FFAppState().cmbStatus == 3) {
        FFAppState().loss = (audit.actweight * audit.marketRate).round().toString();
      }
      FFAppState().flgIrr = '0';
    } else {
      _recalculateTotals();
```

```
      FFAppState().flgIrr = '1';
    }
  });
}

Future<void> _handleTypeChange(int rowIndex, int newTypeId) async {
  final r = FFAppState().Dtview[rowIndex];
  setState(() {
    r.colTypid = newTypeId;
    r.colPurity = '';
    _purityOptionsList[rowIndex] = [];
  });

  final api = ApiService();
  final newOptions = await api.fetchPurityParams(
      FFAppState().pledgeNumIrr,
      FFAppState().selectbrid,
      newTypeId,
      audit.scheme,
      audit.pledgeDate);

  setState(() => _purityOptionsList[rowIndex] = newOptions);
  _dtviewCellEndEdit(rowIndex, 6);
}

void _recalculate() {
  final rows = FFAppState().Dtview;
  final double lndRate = (audit.lendRate ?? 0).toDouble();
  final double pBal = (audit.pledgeBalance ?? 0).toDouble();

  double stoneSum = 0.0;
  double gwSum = 0.0;
  double ntSum = 0.0;
  int totilv = 0;
  int totialv = 0;

  for (var i = 0; i < rows.length; i++) {
    final r = rows[i];
    final double st = double.tryParse(r.colStwt ?? '') ?? 0.0;
    final double gw = double.tryParse(r.colGw ?? '') ?? 0.0;
    final double net = double.tryParse(r.colNetwt ?? '') ?? 0.0;
    final double cnt = (r.colCntwt ?? 0.0).toDouble();

    stoneSum += st;
    gwSum += gw;
    ntSum += net;

    final int ilv = (cnt * lndRate).round();
    final int ialv = (net * lndRate).round();

    totilv += ilv;
    totialv += ialv;

    r.colAutoloss = (ilv - ialv).toString();
  }

  FFAppState().totalStoneVar = double.parse(stoneSum.toStringAsFixed(3));
  FFAppState().totalGwVar = gwSum;
  FFAppState().totalNetWtVar = double.parse(ntSum.toStringAsFixed(3));

  final int totLoanValue = (audit.netweight * lndRate).round();
  final int aalv = totLoanValue - totilv;
  final int totaalv = aalv + totialv;

  final double loss = (pBal > totaalv) ? (pBal - totaalv) : 0.0;
  FFAppState().update(() {
    FFAppState().txtLoss = loss;
  });
}

void _recalculateTotals() {
  double stoneSum = 0, gwSum = 0, ntSum = 0;
  int totilv = 0, totialv = 0;
  double totLoanValue = audit.netweight * audit.lendRate;

  for (final r in FFAppState().Dtview) {
    double st = double.tryParse(r.colStwt ?? '0') ?? 0.0;
    double gw = double.tryParse(r.colGw ?? '0') ?? 0.0;
    double nt = double.tryParse(r.colNetwt ?? '0') ?? 0.0;
    double cnt = r.colCntwt ?? 0.0;

    stoneSum += st;
    gwSum += gw;
    ntSum += nt;
```

```
      if (cnt > 0 && nt >= 0) {
        int ilv = 0;
        int ialv = 0;

        if ([36, 37, 38].contains(FFAppState().cmbStatus)) {
          double rawIlv = cnt * audit.lendRate;
          double rounded2 = double.parse(rawIlv.toStringAsFixed(2));
          ilv = rounded2.round();
          ialv = (nt * audit.lendRate).round();
        } else {
          ilv = (cnt * audit.lendRate).round();
          ialv = (nt * audit.lendRate).round();
        }

        r.colAutoloss = (ilv - ialv).toString();
        totilv += ilv;
        totialv += ialv;
      }
    }

    double totalStone = double.parse(stoneSum.toStringAsFixed(3));
    double totalGw = double.parse(gwSum.toStringAsFixed(3));
    double totalNetWt = double.parse(ntSum.toStringAsFixed(3));

    double finalLoss = _calculateLossFromTotals(
        totLoanValue, totilv, totialv, totalGw, totalStone);

    FFAppState().update(() {
      FFAppState().totalNetWtVar = double.parse(totalNetWt.toStringAsFixed(2));
      FFAppState().totalGwVar = totalGw;
      FFAppState().totalStoneVar = totalStone;
      FFAppState().txtLoss = finalLoss;
    });
  }

  double _calculateLossFromTotals(double totLoanValue, int totilv, int totialv,
      double totalGw, double totalStone) {
    double loss = 0.0;
    if (FFAppState().cmbStatus == 4) {
      loss = ((totalGw - totalStone) * audit.marketRate).roundToDouble();
    } else if ([36, 37, 38].contains(FFAppState().cmbStatus)) {
      final aalv = (totLoanValue - totilv).round();
      final adjustedAalv = (aalv == -1) ? 0 : aalv;
      final totaalv = adjustedAalv + totialv;
      double rawLoss = audit.pledgeBalance - totaalv.toDouble();
      loss = (rawLoss <= -1) ? 0 : rawLoss;
    } else if ([5, 20, 6, 29, 30, 12, 13, 27, 28, 2, 1]
            .contains(FFAppState().cmbStatus) &&
        FFAppState().cmbStatus2 == 0) {
      final aalv = (totLoanValue - totilv).round();
      final totaalv = aalv + totialv;
      loss = (audit.pledgeBalance > totaalv)
          ? (audit.pledgeBalance - totaalv).toDouble()
          : 0.0;
    }
    return loss;
  }

  bool _isStonePurity_Deductible() {
    final dt = DateTime(2013, 2, 8);
    final dt1 = DateTime(2012, 3, 27);
    return audit.pledgeDate.isAfter(dt) || audit.pledgeDate.isBefore(dt1);
  }

  Future<String> _stonePurityDeduction(
      String spurity, String grwt, String stwt, String itemname) async {
    double ded = double.tryParse(spurity) ?? 0.0;
    double g = double.tryParse(grwt) ?? 0;
    double s = double.tryParse(stwt) ?? 0;
    if (g <= 0) return "0";
    double net = g - s;
    double stone = (s / g) * 100;

    if (stone > 85 && itemname.toUpperCase() == "RAJPUTI") {
      _showMessage(
          "Rajaputi Item with stone weight greater than 85% of gross weight is not allowed. Ref : Circular :
5459");
    } else if (stone > 60 && itemname.toUpperCase() != "RAJPUTI") {
      _showMessage(
          "Items with stone weight greater than 50% of gross weight is not allowed. Ref : Circular : 4304");
    }

    double totalDed;
```

```
    double? apiDed = await ApiService().fetchStoneDeduction(audit.pledgeDate, stone);

    if (apiDed != null) {
      double extraDed = apiDed;
      if (stone > 60 && stone <= 85 && itemname.toUpperCase() == "RAJPUTI") {
        extraDed = 5.0;
      }
      totalDed = ded + extraDed;
    } else {
      totalDed = 0.0;
    }

    double sntwt = net - (net * totalDed / 100);
    double roundedSntwt = double.parse(sntwt.toStringAsFixed(2));

    return roundedSntwt < 0 ? "0" : roundedSntwt.toString();
  }

  Future<String> _alcStonePurityDeduction(
      String ntwt, String grwt, String stwt, String itemname) async {
    double g = double.tryParse(grwt) ?? 0;
    double s = double.tryParse(stwt) ?? 0;
    if (g <= 0) return ntwt;
    double net = g - s;
    double stone = (s / g) * 100;

    if (stone > 85 && itemname.toUpperCase() == "RAJPUTI") {
      _showMessage(
          "Rajaputi Item with stone weight greater than 85% of gross weight is not allowed. Ref : Circular :
5459");
    } else if (stone > 60 && itemname.toUpperCase() != "RAJPUTI") {
      _showMessage(
          "Items with stone weight greater than 50% of gross weight is not allowed. Ref : Circular : 4304");
    }

    double ded = 0.0;
    double? apiDed = await ApiService().fetchStoneDeduction(audit.pledgeDate, stone);

    if (apiDed != null) {
      if (stone > 60 && stone <= 85 && itemname.toUpperCase() == "RAJPUTI") {
        ded = 5.0;
      } else {
        ded = apiDed;
      }
    }

    double deductionAmount = double.parse((net * ded / 100).toStringAsFixed(2));
    double currentNtwt = double.tryParse(ntwt) ?? 0;
    double sntwt = double.parse((currentNtwt - deductionAmount).toStringAsFixed(2));

    return sntwt < 0 ? "0" : sntwt.toString();
  }

  Widget _cellNum({
    required double value,
    required ValueChanged<double> onChanged,
    required VoidCallback onSubmitted,
  }) {
    return _NumberCell(
        value: value, onChanged: onChanged, onSubmitted: onSubmitted);
  }

  Widget _cellTxt({
    required String value,
    required ValueChanged<String> onChanged,
    required VoidCallback onSubmitted,
  }) {
    return _TextCell(
        value: value, onChanged: onChanged, onSubmitted: onSubmitted);
  }

  @override
  Widget build(BuildContext context) {
    final autoLossStatuses = [-1, 0, 3, 7, 15, 16, 17, 18, 19, 21, 24, 25, 26, 32];
    final isGridMode = !autoLossStatuses.contains(FFAppState().cmbStatus);

    String gwLabel = 'Gross Wt';
    String purityLabel = 'Purity';
    String estLossLabel = 'Purity Deduction in %';
    bool showPurity = true;
    bool lossreadOnly = true;

    if (FFAppState().cmbStatus == 2) gwLabel = 'Spurious G.W';
    if ([5, 6, 14].contains(FFAppState().cmbStatus))
```

```
    purityLabel = 'Estimated Purity';
  if (FFAppState().cmbStatus == 10) estLossLabel = 'Deduction in LoanAmount';
  if ([23, 9, 10, 4].contains(FFAppState().cmbStatus)) showPurity = false;

  if (_isLoading)
    return const Scaffold(
        backgroundColor: Color(0xFFF5F7FA),
        body: Center(child: CircularProgressIndicator()));

  return Scaffold(
    backgroundColor: const Color(0xFFF5F7FA),
    body: SingleChildScrollView(
      child: Column(
        children: [
          Padding(
            padding: const EdgeInsets.all(16.0),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.stretch,
              children: [
                Card(
                  elevation: 4,
                  shadowColor: Colors.black12,
                  color: Colors.white,
                  shape: RoundedRectangleBorder(
                      borderRadius: BorderRadius.circular(16)),
                  child: Padding(
                    padding: const EdgeInsets.all(20.0),
                    child: Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text('Audit Status',
                            style: Theme.of(context).textTheme.titleMedium),
                        const SizedBox(height: 12),
                        Row(
                          children: [
                            Expanded(
                              child: DropdownButtonFormField<int>(
                                decoration: const InputDecoration(
                                  labelText: 'First irregularity',
                                  filled: true,
                                  fillColor: Colors.white,
                                  border: OutlineInputBorder(),
                                ),
                                value: _statusOptions
                                        .containsKey(FFAppState().cmbStatus)
                                    ? FFAppState().cmbStatus
                                    : null,
                                items: _statusOptions.entries
                                    .map((e) => DropdownMenuItem(
                                        value: e.key,
                                        child: Text(e.value,
                                            overflow: TextOverflow.ellipsis)))
                                    .toList(),
                                onChanged: isEnabled ? _handleStatusChange : null,
                              ),
                            ),
                            const SizedBox(width: 16),
                            Expanded(
                              child: Column(
                                children: [
                                  Container(
                                    padding: const EdgeInsets.all(8),
                                    decoration: BoxDecoration(
                                      color: Colors.white,
                                      borderRadius: BorderRadius.circular(8),
                                      border: Border.all(color: Colors.black),
                                    ),
                                    width: double.infinity,
                                    height: 48,
                                    child: Center(
                                      child: Text(
                                        'First irregularity: ${((FFAppState().cmbStatus != 0) ?
_statusOptions[FFAppState().cmbStatus] ?? '' : '')}',
                                        style: Theme.of(context)
                                            .textTheme
                                            .bodyMedium!
                                            .copyWith(
                                                fontWeight: FontWeight.w500,
                                                color: Colors.red),
                                      ),
                                    ),
                                  )
                                ],
                              ),
```

```
                    ),
                    const SizedBox(width: 16),
                    Expanded(
                      child: DropdownButtonFormField<int>(
                        decoration: const InputDecoration(
                          labelText: 'Second irregularity',
                          filled: true,
                          fillColor: Colors.white,
                          border: OutlineInputBorder(),
                        ),
                        value: _statusOptions
                                .containsKey(FFAppState().cmbStatus2)
                            ? FFAppState().cmbStatus2
                            : null,
                        items: _statusOptions.entries
                            .map((e) => DropdownMenuItem(
                                value: e.key,
                                child: Text(e.value,
                                    overflow: TextOverflow.ellipsis)))
                            .toList(),
                        onChanged: isEnabled
                            ? (v) {
                                setState(() {
                                  FFAppState().cmbStatus2 =
                                      v ?? FFAppState().cmbStatus2;
                                  if (isGridMode) _recalculateTotals();
                                });
                              }
                            : null,
                      ),
                    ),
                  ],
                ),
                if (FFAppState().cmbStatus == 3) ...[
                  const SizedBox(height: 16),
                  SizedBox(
                    width: 400,
                    child: TextFormField(
                      readOnly: lossreadOnly,
                      decoration: InputDecoration(
                        border: const OutlineInputBorder(),
                        hintText: 'Loss Amount: ${FFAppState().loss}',
                      ),
                      keyboardType: const TextInputType.numberWithOptions(decimal: true),
                      onChanged: (v) {
                        setState(() {
                          FFAppState().update(() {
                            FFAppState().txtLoss = double.tryParse(v) ?? 0.0;
                          });
                        });
                      },
                    ),
                  ),
                ],
                if (FFAppState().cmbStatus == 32) ...[
                  const SizedBox(height: 16),
                  SizedBox(
                    width: 400,
                    child: TextFormField(
                      decoration: InputDecoration(
                        hintText: audit.tareWeight == 0
                            ? 'Enter tare weight'
                            : audit.tareWeight.toString(),
                        border: const OutlineInputBorder(),
                      ),
                      initialValue: audit.tareWeight == 0
                          ? ''
                          : audit.tareWeight.toString(),
                      keyboardType: const TextInputType.numberWithOptions(decimal: true),
                      onChanged: (v) {
                        setState(() {
                          audit.tareWeight = double.tryParse(v) ?? 0.0;
                          FFAppState().update(() {
                            FFAppState().newtareweight = v;
                          });
                        });
                      },
                    ),
                  ),
                ],
              ],
            ),
          ),
        ),
```

```
                if (isGridMode)
                  Card(
                    elevation: 8,
                    shadowColor: const Color(0xFF667eea).withOpacity(0.4),
                    color: Colors.white,
                    shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(20)),
                    child: Padding(
                      padding: const EdgeInsets.all(24.0),
                      child: Column(
                        children: [
                          Row(
                            mainAxisAlignment: MainAxisAlignment.spaceBetween,
                            children: [
                              _SummaryItem(
                                  label: 'Total GW',
                                  value: FFAppState().totalGwVar.toStringAsFixed(3)),
                              _SummaryItem(
                                  label: 'Total Stone',
                                  value: FFAppState().totalStoneVar.toStringAsFixed(3)),
                              _SummaryItem(
                                  label: 'Total Net',
                                  value: FFAppState().totalNetWtVar.toStringAsFixed(3),
                                  isBold: true),
                            ],
                          ),
                          const Divider(height: 32),
                          Row(
                            mainAxisAlignment: MainAxisAlignment.center,
                            children: [
                              Text('Estimated Loss: ',
                                  style: Theme.of(context).textTheme.titleMedium),
                              Text(
                                FFAppState().txtLoss.toStringAsFixed(2),
                                style: Theme.of(context).textTheme.headlineSmall?.copyWith(
                                      color: Colors.black,
                                      fontWeight: FontWeight.bold,
                                    ),
                              ),
                            ],
                          ),
                        ],
                      ),
                    ),
                  ),
              ],
            ),
          ),
          if (isGridMode) ...[
            Card(
              margin: const EdgeInsets.symmetric(horizontal: 16),
              elevation: 8,
              shadowColor: const Color(0xFF667eea).withOpacity(0.25),
              clipBehavior: Clip.antiAlias,
              color: Colors.white,
              shape: RoundedRectangleBorder(
                  borderRadius: BorderRadius.circular(16)),
              child: SingleChildScrollView(
                scrollDirection: Axis.horizontal,
                child: DataTable(
                  columnSpacing: 20,
                  headingRowColor:
                      MaterialStateProperty.all(const Color(0xFF667eea)),
                  headingTextStyle: const TextStyle(
                      fontWeight: FontWeight.bold, color: Colors.white),
                  columns: [
                    const DataColumn(label: Text('Item Name', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 11))),
                    const DataColumn(label: Text('Select', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 11))),
                    const DataColumn(label: Text('No of Pieces', style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 11))),
                    DataColumn(label: Text(gwLabel)),
                    const DataColumn(label: Text('Net Wt', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 11))),
                    const DataColumn(label: Text('Stone Wt', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 11))),
                    const DataColumn(label: Text('CurrentPurity', style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 11))),
                    if (showPurity) DataColumn(label: Text('Item Type', style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 11))),
                    if (showPurity) DataColumn(label: Text(purityLabel, style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 11))),
                    if (showPurity) DataColumn(label: Text(estLossLabel, style: TextStyle(fontWeight:
```

```
FontWeight.bold, fontSize: 11))),
                    ],
                    rows: List.generate(FFAppState().Dtview.length, (i) {
                      final r = FFAppState().Dtview[i];
                      final purityOptions = _purityOptionsList.length > i
                          ? _purityOptionsList[i]
                          : <String>[];
                      return DataRow(
                        cells: [
                          DataCell(SizedBox(
                              width: 120,
                              child: Text(r.colItem ?? '',
                                  style: const TextStyle(
                                      fontWeight: FontWeight.w500,
                                      fontSize: 11)))),
                          DataCell(Checkbox(
                            value: r.colStatus,
                            onChanged: (v) {
                              setState(() => r.colStatus = v ?? false);
                              _dtviewCellEndEdit(i, 1);
                            },
                          )),
                          DataCell(_cellNum(
                            value: double.tryParse(r.colItemcount ?? '0') ?? 0.0,
                            onChanged: (v) {
                              r.colItemcount = v.toInt().toString();
                              _dtviewCellEndEdit(i, 2);
                            },
                            onSubmitted: () => _dtviewCellEndEdit(i, 2),
                          )),
                          DataCell(_cellNum(
                            value: double.tryParse(r.colGw ?? '0') ?? 0.0,
                            onChanged: (v) {
                              r.colGw = v.toString();
                              _dtviewCellEndEdit(i, 3);
                            },
                            onSubmitted: () => _dtviewCellEndEdit(i, 3),
                          )),
                          DataCell(Text(
                              (double.tryParse(r.colNetwt ?? '0') ?? 0.0).toStringAsFixed(3),
                              style: const TextStyle(
                                  fontWeight: FontWeight.bold, fontSize: 11))),
                          DataCell(_cellNum(
                            value: double.tryParse(r.colStwt ?? '0') ?? 0.0,
                            onChanged: (v) {
                              r.colStwt = v.toString();
                              _dtviewCellEndEdit(i, 5);
                            },
                            onSubmitted: () => _dtviewCellEndEdit(i, 5),
                          )),
                          DataCell(SizedBox(
                              width: 120,
                              child: Text(r.colCurpurity ?? '',
                                  style: const TextStyle(
                                      fontWeight: FontWeight.w500,
                                      fontSize: 11)))),
                          if (showPurity)
                            DataCell(SizedBox(
                              width: 160,
                              child: _itemTypes.isNotEmpty
                                  ? DropdownButtonFormField<int>(
                                      hint: const Text('Select Type', style: TextStyle(fontSize: 11)),
                                      value: _itemTypes.containsKey(r.colTypid) ? r.colTypid : null,
                                      isExpanded: true,
                                      decoration: const InputDecoration(border: InputBorder.none,
contentPadding: EdgeInsets.zero),
                                      items: _itemTypes.entries
                                          .map((e) => DropdownMenuItem(
                                                value: e.key,
                                                child: Text(e.value, overflow: TextOverflow.ellipsis, style:
const TextStyle(fontSize: 11)),
                                              ))
                                          .toList(),
                                      onChanged: (v) {
                                        if (v != null) _handleTypeChange(i, v);
                                      },
                                    )
                                  : Text(_itemTypes[r.colTypid] ?? ''),
                            )),
                          if (showPurity)
                            DataCell(SizedBox(
                              width: 160,
                              child: purityOptions.isNotEmpty
                                  ? DropdownButtonFormField<String>(
```

```
                                    hint: const Text('Select Purity', style: TextStyle(fontSize: 11)),
                                    value: purityOptions.contains(r.colPurity) ? r.colPurity : null,
                                    isExpanded: true,
                                    decoration: const InputDecoration(border: InputBorder.none,
contentPadding: EdgeInsets.zero),
                                    items: purityOptions
                                        .map((e) => DropdownMenuItem(
                                              value: e,
                                              child: Text(e.split('~').first, overflow:
TextOverflow.ellipsis, style: const TextStyle(fontSize: 11)),
                                            ))
                                        .toList(),
                                    onChanged: (v) {
                                      if (v != null) {
                                        r.colPurity = v;
                                        _dtviewCellEndEdit(i, 8);
                                      }
                                    },
                                  )
                                : _cellTxt(
                                    value: r.colPurity ?? '',
                                    onChanged: (v) {
                                      r.colPurity = v;
                                      _dtviewCellEndEdit(i, 8);
                                    },
                                    onSubmitted: () => _dtviewCellEndEdit(i, 8),
                                  ),
                          )),
                        if (showPurity)
                          DataCell(Text(
                              (double.tryParse(r.colEstimat ?? '0') ?? 0.0).toStringAsFixed(0),
                              style: const TextStyle(fontWeight: FontWeight.bold, fontSize: 11))),
                      ],
                    );
                  }),
                ),
              ),
            ),
          ] else
            const Center(
              child: Column(
                children: [
                  SizedBox(height: 40),
                ],
              ),
            ),
          const SizedBox(height: 80),
        ],
      ),
    ),
  );
 }
}

class _SummaryItem extends StatelessWidget {
  final String label;
  final String value;
  final bool isBold;
  final Color? textColor;
  const _SummaryItem({
    required this.label,
    required this.value,
    this.isBold = false,
    this.textColor,
  });

  @override
  Widget build(BuildContext context) {
    return Column(
      children: [
        Text(label,
            style: Theme.of(context)
                .textTheme
                .bodySmall
                ?.copyWith(color: textColor?.withOpacity(0.8))),
        Text(value,
            style: Theme.of(context).textTheme.titleMedium?.copyWith(
                fontWeight: isBold ? FontWeight.bold : FontWeight.normal,
                color: textColor,
                fontSize: 18)),
      ],
    );
  }
}
```

```
class AuditState {
  double lendRate;
  int pledgeBalance;
  double totalNetWtDb;
  String scheme;
  DateTime pledgeDate;
  double marketRate;
  double actweight;
  double netweight;
  int? spotRectificationStatus;
  String spotRemarks = '';
  DateTime rectificationDate = DateTime.now();
  int kycStatus = 0;
  int invStatus = 0;
  bool isGoldBar = false;
  bool isGoldCoin = false;
  String remarks = '';
  String packetIntact = '';
  String auditorNotice = '';
  String prevAuditor = '';
  String multipleIds = '';
  int recoveryProspect = -1;
  String firNo = '';
  String blobPath = '';
  double tareWeight = 0.0;

  AuditState({
    required this.lendRate,
    required this.pledgeBalance,
    required this.totalNetWtDb,
    required this.scheme,
    required this.pledgeDate,
    required this.actweight,
    required this.netweight,
    this.marketRate = 0.0,
  });
}

typedef ShowMessage = void Function(String message);

class AlcInput {
  final String itemName;
  final double deduRate;
  final int dbItemCount;
  final int itemId;
  final int brid;
  final String pledgeNo;
  final double cgwt;
  final double cstwt;
  final int statusMain;
  final double cntwt;
  final String scheme;
  final DateTime pledgeDate;
  final int statusSub;
  final int typeId;
  final String curPurity;
  final String itemType;
  final double lendRate;

  AlcInput({
    required this.itemName,
    required this.deduRate,
    required this.dbItemCount,
    required this.itemId,
    required this.brid,
    required this.pledgeNo,
    required this.cgwt,
    required this.cstwt,
    required this.statusMain,
    required this.cntwt,
    required this.scheme,
    required this.pledgeDate,
    required this.statusSub,
    required this.typeId,
    required this.curPurity,
    required this.itemType,
    required this.lendRate,
  });
}

class AlcResult {
  final double tGrwt, tStwt, tNtwt;
  final String subrecord;
```

```
  AlcResult({
    required this.tGrwt,
    required this.tStwt,
    required this.tNtwt,
    required this.subrecord,
  });
}

typedef OpenAlcForm = Future<AlcResult?> Function(AlcInput input);

bool isStonePurityDeductible(DateTime pledgeDate) {
  final start = DateTime(2013, 2, 8), end = DateTime(2012, 3, 27);
  return pledgeDate.isAfter(start) || pledgeDate.isBefore(end);
}

Future<double> stonePurityDeduction({
  required double spurityPercent,
  required double grwt,
  required double stwt,
  required String itemName,
  required DateTime pledgeDate,
  required ShowMessage showMessage,
  Future<double?> Function(double stonePct, DateTime pledgeDate)? stoneDeductionLookup,
}) {
  final g = grwt;
  if (g <= 0) return Future.value(0.0);
  final s = stwt;
  return Future(() async {
    final stonePct = (s / g) * 100.0;
    if (stonePct > 85 && itemName.toUpperCase() == 'RAJPUTI') {
      showMessage('Rajaputi item stone >85% not allowed (Circular 5459)');
    } else if (stonePct > 60 && itemName.toUpperCase() != 'RAJPUTI') {
      showMessage('Items stone >50% not allowed (Circular 4304)');
    }
    double? dedExtra;
    if (stoneDeductionLookup != null) {
      dedExtra = await stoneDeductionLookup(stonePct, pledgeDate);
    }

    double totalDed = 0.0;
    if (dedExtra != null) {
      double extra = dedExtra;
      if (stonePct > 60 && stonePct <= 85 && itemName.toUpperCase() == 'RAJPUTI') extra = 5.0;
      totalDed = max(0.0, spurityPercent + extra);
    }

    final netGold = max(0.0, g - s);
    final sntwt = (netGold - (netGold * totalDed / 100.0));
    return sntwt <= 0 ? 0.0 : double.parse(sntwt.toStringAsFixed(2));
  });
}

Future<double> alcStonePurityDeduction({
  required double ntwt,
  required double grwt,
  required double stwt,
  required String itemName,
  required DateTime pledgeDate,
  required ShowMessage showMessage,
  Future<double?> Function(double stonePct, DateTime pledgeDate)? stoneDeductionLookup,
}) {
  final g = grwt;
  if (g <= 0) return Future.value(0.0);
  final s = stwt;
  return Future(() async {
    final stonePct = (s / g) * 100.0;
    if (stonePct > 85 && itemName.toUpperCase() == 'RAJPUTI') {
      showMessage('Rajaputi item stone >85% not allowed (Circular 5459)');
    } else if (stonePct > 60 && itemName.toUpperCase() != 'RAJPUTI') {
      showMessage('Items stone >50% not allowed (Circular 4304)');
    }
    double? dedExtra;
    if (stoneDeductionLookup != null) {
      dedExtra = await stoneDeductionLookup(stonePct, pledgeDate);
    }

    double dedVal = 0.0;
    if (dedExtra != null) {
      if (stonePct > 60 && stonePct <= 85 && itemName.toUpperCase() == 'RAJPUTI') {
        dedVal = 5.0;
      } else {
        dedVal = dedExtra;
      }
    }
```

```
    final netGold = max(0.0, g - s);
    final stoneDed = netGold * (max(0.0, dedVal) / 100.0);
    final val = ntwt - double.parse(stoneDed.toStringAsFixed(2));
    return val <= 0 ? 0.0 : double.parse(val.toStringAsFixed(2));
  });
}

class _NumberCell extends StatefulWidget {
  final double value;
  final ValueChanged<double> onChanged;
  final VoidCallback onSubmitted;

  const _NumberCell({
    required this.value,
    required this.onChanged,
    required this.onSubmitted,
  });

  @override
  State<_NumberCell> createState() => _NumberCellState();
}

class _NumberCellState extends State<_NumberCell> {
  late TextEditingController _ctrl;

  @override
  void initState() {
    super.initState();
    _ctrl = TextEditingController(text: widget.value == 0 ? '' : widget.value.toString());
  }

  @override
  void didUpdateWidget(_NumberCell oldWidget) {
    super.didUpdateWidget(oldWidget);
    if (widget.value != oldWidget.value) {
      final currentVal = double.tryParse(_ctrl.text) ?? 0.0;
      if ((currentVal - widget.value).abs() > 0.0001) {
        _ctrl.text = widget.value == 0 ? '' : widget.value.toString();
        _ctrl.selection = TextSelection.collapsed(offset: _ctrl.text.length);
      }
    }
  }

  @override
  void dispose() {
    _ctrl.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return SizedBox(
      width: 100,
      child: TextField(
        controller: _ctrl,
        keyboardType: TextInputType.number,
        onChanged: (v) => widget.onChanged(double.tryParse(v.trim()) ?? 0.0),
        onSubmitted: (_) => widget.onSubmitted(),
        decoration: const InputDecoration(isDense: true, border: OutlineInputBorder()),
      ),
    );
  }
}

class ApiService {
  static String baseUrl = FFDevEnvironmentValues().Baseurl + '/AuditInspectionApi/api/GetInspectionData';
  String token = FFAppState().Token;

  Future<List<dynamic>> _get(String flag, String params) async {
    final url = Uri.parse('$baseUrl/$flag/$params/1');
    try {
      final response = await http.get(
        url,
        headers: {'Authorization': 'Bearer $token'},
      );
      if (response.statusCode == 200) {
        final data = jsonDecode(response.body);
        if (data is Map && data.containsKey('result')) {
          return data['result'] as List<dynamic>;
        }
      } else {
        debugPrint('API Error ${response.statusCode}: ${response.body}');
      }
```

```
    } catch (e) {
      debugPrint('API Exception: $e');
    }
    return [];
  }

  Future<List<Result1Struct>> fetchPledgeItems(String pledgeNo, String brid) async {
    final data = await _get('IRREGULARITY_VIEW', '$pledgeNo*${FFAppState().cmbStatus}*$brid');
    return data.map((json) {
      String itemName = '';
      json.forEach((k, v) {
        if (k.toString().contains('ITEM_NAME')) itemName = v.toString().trim();
        if (k.toString().contains('RPAD(IM.ITEM_NAME,30)')) itemName = v.toString().trim();
      });
      return Result1Struct(
        colItem: itemName,
        colItemid: int.tryParse(json['ITEM_ID']?.toString() ?? '0') ?? 0,
        colCurpurity: json['DEDUCTION_NAME']?.toString() ?? '',
        colDeducrate: int.tryParse(json['DEDUCTION_RATE']?.toString().split('~').last ?? '0') ?? 0,
        colAutoloss: '0',
        colDbitemcount: int.tryParse(json['ITEM_COUNT']?.toString() ?? '0') ?? 0,
        colTypid: int.tryParse(json['TYPE_ID']?.toString() ?? '0') ?? 0,
        colCgwt: double.tryParse(json['ACT_WT']?.toString() ?? '0') ?? 0.0,
        colCstwt: int.tryParse(json['STONE_WT']?.toString() ?? '0') ?? 0,
        colCntwt: double.tryParse(json['NET_WT']?.toString() ?? '0') ?? 0.0,
        colPurity: json['DEDUCTION_NAME']?.toString() ?? '',
        colStatus: false,
        colItemcount: '0',
        colGw: '0',
        colStwt: '0',
        colNetwt: '0',
        colEstimat: '0',
        colCirNo: '',
        colSubrecord: '',
        colItemtype: 0,
      );
    }).toList();
  }

  Future<double?> fetchStoneDeduction(DateTime date, double stonePct) async {
    const months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'];
    final dateStr = "${date.day.toString().padLeft(2, '0')}-${months[date.month - 1]}-${date.year}";
    final data = await _get('PLP_AUDIT_SELECT', 'STONEDEDUCTION*0*0*$dateStr*$stonePct*0');
    return data.isNotEmpty ? (double.tryParse(data[0]['DED_PERC']?.toString() ?? '0')) : null;
  }

  Future<double> fetchMarketRate() async {
    final data = await _get('PLP_AUDIT_SELECT', 'MARKETRATE*0*0*0*0*0*0');
    return data.isNotEmpty ? (double.tryParse(data[0]['PARMTR_VALUE'].toString()) ?? 0.0) : 0.0;
  }

  Future<double> fetchLendRate(String pledgeNo) async {
    final data = await _get('PLP_AUDIT_SELECT', 'LENDRATE*$pledgeNo*0*0*0*0*0');
    return data.isNotEmpty ? (double.tryParse(data[0]['LND_RATE'].toString()) ?? 0.0) : 0.0;
  }

  Future<int> fetchPledgeBalance(String pledgeNo) async {
    final data = await _get('LOAD_PLEDGE_BALANCE', pledgeNo);
    if (data.isEmpty) return 0;
    final val = data[0]['PLEDGE_BAL'];
    return val != null ? (double.tryParse(val.toString())?.toInt() ?? 0) : 0;
  }

  Future<List<String>> fetchPurityParams(String pledgeNo, String brid, int itemType, String scheme, DateTime
date) async {
    const months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'];
    final dateStr = "${date.day.toString().padLeft(2, '0')}-${months[date.month - 1]}-${date.year}";
    final params = 'PURITYPARAMS*$pledgeNo*$brid*$itemType*$scheme*$dateStr';
    final data = await _get('PLP_AUDIT_SELECT', params);
    return data.map((e) => e['DEDUCTION_RATE'].toString()).toList();
  }

  Future<Map<int, String>> fetchItemTypes() async {
    final data = await _get('PLP_AUDIT_SELECT', 'ITEMTYPES*0*0*0*0*0*0');
    final options = <int, String>{};
    for (final item in data) {
      final id = item['TYPE_ID'];
      final name = item['TYPE_NAME'];
      if (id != null && name != null) {
        options[int.parse(id.toString())] = name.toString();
      }
    }
    return options;
  }
```

```
  Future<Map<int, String>> fetchStatusOptions(String brid, int stflg) async {
    final data = await _get('STATEID_CHECK', '$brid*$stflg');
    final options = <int, String>{};
    for (final item in data) {
      var id;
      var name;

      if (item is Map) {
        for (var key in item.keys) {
          if (key == '0' || key == 'STATUS_ID') id = item[key];
          if (key == "'-------SELECT---------'" || key == 'STATUS_NAME') name = item[key];
        }
        if (id == null && item.values.isNotEmpty) {
          final vals = item.values.toList();
          if (vals.isNotEmpty) id = vals[0];
          if (vals.length > 1) name = vals[1];
        }
      }

      if (id != null && name != null) {
        final parsedId = double.tryParse(id.toString())?.toInt();
        if (parsedId != null) {
          options[parsedId] = name.toString();
        }
      }
    }
    return options;
  }

  Future<Map<String, dynamic>?> fetchInventoryDetails(String brid, String pledgeNo) async {
    final data = await _get('LOAD_INV_DETAILS', '$brid*$pledgeNo');
    if (data.isNotEmpty) return data[0];
    return null;
  }
}

class _TextCell extends StatefulWidget {
  final String value;
  final ValueChanged<String> onChanged;
  final VoidCallback onSubmitted;

  const _TextCell({
    required this.value,
    required this.onChanged,
    required this.onSubmitted,
  });

  @override
  State<_TextCell> createState() => _TextCellState();
}

class _TextCellState extends State<_TextCell> {
  late TextEditingController _ctrl;

  @override
  void initState() {
    super.initState();
    _ctrl = TextEditingController(text: widget.value);
  }

  @override
  void didUpdateWidget(_TextCell oldWidget) {
    super.didUpdateWidget(oldWidget);
    if (widget.value != oldWidget.value && widget.value != _ctrl.text) {
      _ctrl.text = widget.value;
      _ctrl.selection = TextSelection.collapsed(offset: _ctrl.text.length);
    }
  }

  @override
  void dispose() {
    _ctrl.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return SizedBox(
      width: 160,
      child: TextField(
        controller: _ctrl,
        onChanged: widget.onChanged,
        onSubmitted: (_) => widget.onSubmitted(),
```

```
        decoration: const InputDecoration(isDense: true, border: OutlineInputBorder()),
      ),
    );
  }
}
```